



Molecular Genetics and Inheritance

Name: _____

Class: _____

1. A sample of double-stranded DNA was analysed in a laboratory. It was found that 32% of the nitrogenous bases present were thymine (T). What is the percentage of guanine (G) in this DNA sample?

- A. 18%
- B. 32%
- C. 36%
- D. 68%

2. Which of the following statements correctly describes the molecular structure of DNA established by Watson and Crick?

- A. The two sugar-phosphate chains run antiparallel to each other.
- B. Nitrogenous bases form covalent bonds along the same chain to link adjacent nucleotides.
- C. The sugar constituent alternates between ribose and deoxyribose along the chain.
- D. Four different types of phosphate groups are present in equal proportions.

3. A pedigree study was conducted on a family with a history of sickle-cell anaemia. Genetic testing revealed that sickle-cell anaemia is caused by a base substitution in the beta-globin gene. How does this gene mutation lead to the sickling of red blood cells under low oxygen conditions?

- A. An altered nucleotide sequence results in a modified amino acid sequence, changing the structure of haemoglobin.
- B. An extra autosome disrupts normal protein synthesis during erythropoiesis.
- C. A large deletion of chromosomal material removes the gene responsible for membrane flexibility.
- D. Translocation of genes between non-homologous chromosomes prevents haemoglobin maturation.

4. A pregnant woman undergoes amniocentesis for foetal diagnostic screening. The isolated foetal cells are cultured and stopped at metaphase to prepare a karyotype.

Which of the following conditions can be directly diagnosed from this karyotype analysis?

- I. Down syndrome
- II. Sickle-cell anaemia
- III. Klinefelter syndrome (XXY condition)

- A. I and II only
- B. I and III only
- C. II and III only
- D. I, II and III

5. The Human Genome Project (HGP) successfully determined the standard reference sequence of human DNA. Despite this achievement, why is reference genome sequence data alone insufficient to precisely predict an individual's susceptibility to complex medical conditions such as hypertension or type 2 diabetes?

- A. Most complex human health traits are polygenic and heavily influenced by environmental and lifestyle factors.
- B. The Human Genome Project mapped only mRNA sequences rather than genomic DNA sequences.
- C. Individual human cells contain completely different sets of genes depending on the tissue type.
- D. DNA sequence data provides information about nuclear inheritance but completely excludes mitochondrial genes.

6. Which of the following statements highlights an ethical concern directly associated with the widespread application of genomic screening data derived from the Human Genome Project?

- A. Potential misuse of personal genetic data by third parties leading to insurance or employment discrimination.
- B. The permanent modification of essential human metabolic enzymes caused by routine gene sequencing.
- C. The total loss of natural selection due to accurate mapping of autosomal gene loci.
- D. An inevitable increase in spontaneous gene mutation rates across the general population.

7. A man with blood group A and a woman with blood group B have a child who has blood group O. What is the probability that their next child will have blood group B?

- A. 0
- B. 0.25
- C. 0.50
- D. 0.75

8. In a maternity ward, four babies were born with different ABO blood groups. Which of the following parental combinations could NOT possibly give rise to a child with blood group O?

- A. Group A father × Group B mother
- B. Group A father × Group A mother
- C. Group AB father × Group O mother
- D. Group B father × Group O mother

9. Students performed a DNA extraction experiment using kiwi fruit. After mashing the tissue, liquid detergent and protease enzyme solution were added. Cold ethanol was then carefully poured down the side of the test tube.

What is the primary role of adding cold ethanol in this protocol?

- A. To precipitate the insoluble DNA from the aqueous solution so it can be collected.
- B. To dissolve the lipid bilayers of the plasma membrane and nuclear envelope.
- C. To digest histone proteins bound to the double-stranded DNA structure.
- D. To break the hydrogen bonds between complementary base pairs.

10. In an agarose gel electrophoresis experiment to analyse DNA fragments, samples were loaded into wells near the negative electrode (cathode) and an electric current was applied across the gel buffer.

Which row in the table correctly explains why DNA fragments move toward the positive electrode (anode) and how their migration distances compare?

	Charge on DNA responsible for movement	Migration speed of smaller fragments co
A	Negative charge due to phosphate groups	Faster migration distance
B	Negative charge due to deoxyribose sugars	Slower migration distance
C	Positive charge due to nitrogenous bases	Faster migration distance
D	Positive charge due to phosphate groups	Slower migration distance

- A. A
- B. B
- C. C
- D. D

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Answers

1 A

In double-stranded DNA, adenine pairs with thymine ($A = T$) and guanine pairs with cytosine ($G = C$).

1. Given $T = 32\%$, adenine must also be $A = 32\%$.

2. Combined, $A + T = 32\% + 32\% = 64\%$.

3. The remaining bases ($G + C$) account for $100\% - 64\% = 36\%$.

4. Since $G = C$, the percentage of guanine is $\frac{36\%}{2} = 18\%$.

2 A

The DNA model proposed by Watson and Crick consists of two antiparallel polynucleotide strands running in opposite directions, forming a double helix. The sugar-phosphate backbones form the outer structural frame, while nitrogenous bases point inward and pair via hydrogen bonds. The sugar present is deoxyribose, which is uniform throughout DNA, and the phosphate group is identical across all nucleotides.

3 A

A gene mutation involves a alteration in the sequence of nucleotides in DNA. In sickle-cell anaemia, a single base substitution alters a codon in mRNA, leading to a single amino acid substitution in the haemoglobin protein. This altered haemoglobin polymerises under low oxygen tension, distorting the red blood cell into a sickle shape. It does not involve structural changes to whole chromosomes or chromosomal non-disjunction.

4 B

A karyotype is a visual representation of all chromosomes in a cell, organised by size, banding pattern, and centromere position. It is used to detect chromosome mutations (changes in chromosome number or large structural alterations).

1. Down syndrome (trisomy 21) involves an extra chromosome 21 and is clearly visible on a karyotype.
2. Sickle-cell anaemia is a gene mutation (single base pair substitution) and cannot be seen on a karyotype.
3. Klinefelter syndrome (XXY) is a numerical sex chromosome abnormality and is clearly visible on a karyotype.

Therefore, statements I and III can be directly diagnosed.

5 A

Complex diseases such as hypertension and type 2 diabetes are polygenic (controlled by many genes interacting together) and strongly influenced by environmental factors (lifestyle, diet, exercise). Knowing a single reference sequence does not fully account for complex gene-environment interactions, epigenetic variations, or all individual genetic variants across populations.

6 A

Widespread genetic testing creates ethical dilemmas regarding genetic privacy and potential genetic discrimination by employers or insurance companies based on an individual's predisposition to disease. Options B, C, and D represent scientific misunderstandings rather than ethical implications.

In the ABO blood group system, alleles I^A and I^B are codominant, and both are dominant to the recessive allele i .

1. Since the child has blood group O, its genotype must be ii . This means both parents carry the recessive allele i .
2. Therefore, the father's genotype is $I^A i$ and the mother's genotype is $I^B i$.
3. Performing a genetic cross ($I^A i \times I^B i$):

	I^B	i
I^A	$I^A I^B$ (Group AB)	$I^A i$ (Group A)
i	$I^B i$ (Group B)	ii (Group O)

4. The possible offspring genotypes are $I^A I^B$, $I^A i$, $I^B i$, and ii in equal proportions (25% each).
5. The probability of having a child with blood group B ($I^B i$) is $\frac{1}{4}$ or 0.25.

To have a child with blood group O (genotype ii), both parents must pass on a recessive i allele.

A parent with blood group AB has the genotype $I^A I^B$. This parent produces gametes carrying either the I^A allele or the I^B allele, but never the i allele. Thus, any offspring receiving a gamete from an AB parent will inherit either I^A or I^B , making genotype ii (blood group O) impossible.

DNA is soluble in water/aqueous solutions but insoluble in alcohol (ethanol). When ice-cold ethanol is added, DNA precipitates out of solution at the interface between the aqueous layer and ethanol layer, forming visible white threads that can be wound or collected. Detergent breaks down cell membranes, and protease breaks down histone proteins.

1. The phosphate backbone of DNA carries a net negative charge in neutral gel running buffers, causing DNA to migrate toward the positive electrode (anode).
2. The agarose gel acts as a molecular sieve. Smaller DNA fragments experience less resistance moving through the gel matrix pores, so they travel faster and further than larger fragments in a given time.